Sarvesh Patham

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Career Objective

Improve industry experience in robotics, digital hardware design, and physical design through a **summer internship**

Education

University of Illinois Urbana Champaign Urbana, IL

Bachelor of Sciences in Electrical Engineering, Minor in Computer Science *Expected: May 2028*

- Cumulative GPA: 3.9

Experience

Astranis Fall 2026

Hardware Test Engineering Intern *San Francisco, CA*

- Built Python-based Hardware Test Framework (HTF) tests to validate onboard satellite propulsion systems
- Wrote simulations of satellite hardware testbeds to verify test functionality enabling software-level validation
- Updated gyroscope and accelerometer drivers for radiation testing, increasing sampling rates from 15 to 83 Hz

ARM Summer 2026

Physical Implementation Intern *Austin, TX*

- Developed netlist backtracing CLI/GUI tool for detecting integrated clock gates and verifying technology mapping
- Created large RTL hierarchy processing flow to translate hierarchies into easily traversable directed acyclic graphs
- Verified block-level physical designs and parsed bottlenecks utilizing Cadence Innovus and Synopsys Verdi

Electrical Engineer Intern June 2025 – Present

General Autonomy AI *Piscataway, NJ*

- Developed LiDAR/Camera/IMU data collection stack using NVIDIA Jetson Orin for simulator development
- Implemented autonomy stack using ROS 2 capable of lanelet planning and object detection from sensor data
- Secured $50k in invesments through LiDAR and Camera mount optimization for investor demos.

Undergraduate Research Assistant August 2025 – May 2026

University of Illinois Urbana Champaign *Urbana, IL*

- Constructed magnetic field verification system using Python, successfully field tested simulation field lines
- Designed hall effect sensor mount and transceiver + pre-amp housing using Fusion360
- Researching transceiver designs for decreasing signal-to-noise ratio and increasing data clarity for low field MRI

Projects/Extracurriculars

SigRobotics | *Electrical Engineer* February 2026 – Present

- Determined stepper motor specifications, sensors, and gantry design for an industrial use autosampler
- Controlling autonomous sampling system via PLC in C++ for remote access through Ignition
- Designing pinch valves in Fusion360 for tubing based on industry specifications for analog controlled dispense

Shazam Song Matcher | *FPGA/Embedded Project* March 2026 – May 2026

- Developed a Shazam-style matcher that captured live audio and generated frequency hashes in SystemVerilog
- Used MicroBlaze C in Vitis to read BRAM hashes, compare against a database, and output matches over UART
- Built Python tools for UART audio streaming, song recording, and fingerprint database generation

Cuff-Link | *Embedded Systems Project* December 2023 – May 2024

- Designed a muscle and motion tracking device using Fusion360 and developed in the Arduino IDE
- Programmed python based virtual cursor controlled by onboard IMU and Electromyography sensor (EMG)
- Selected as a top 3 class finalist for school competition; demonstrated product to audience of 400+

Technical Skills

Digital: Python, Robotics Operating System(ROS 2), MATLAB, Linux, C,C++, HTML /CSS, CAD (Fusion360/OnShape), Arduino, ESP32

Hardware:PCB Design(KiCAD,Altium), LTspice/SPICE, SystemVerilog, Verilog, VHDL, Oscilloscopes, Function Generators, Soldering, Communcation Protocols (UART/I2C/CAN)